

MODULE *messages*

Message constructors for the *Zcash P2P* wire protocol (*ZIP-0204*).

Each operator builds a record with a header (magic number, command name) and a *payload* whose fields mirror the on-wire format. Fields that are irrelevant to the properties being verified are set to constant values to keep the *TLC* state space manageable.

EXTENDS *Naturals*, *FiniteSets*

$MakeVersion(from, to, blocks) \triangleq [$
 $\quad header \mapsto [magic \mapsto 619259748, command \mapsto "version"],$
 $\quad payload \mapsto [$
 $\quad\quad version \mapsto 170100,$
 $\quad\quad services \mapsto 0,$
 $\quad\quad addr_recv \mapsto to,$
 $\quad\quad addr_from \mapsto from,$
 $\quad\quad nonce \mapsto 0,$
 $\quad\quad user_agent \mapsto "",$
 $\quad\quad start_height \mapsto Cardinality(blocks),$
 $\quad\quad relay \mapsto FALSE$
 $\quad]$
 $]$

$MakeVerack \triangleq [$
 $\quad header \mapsto [magic \mapsto 619259748, command \mapsto "verack"]$
 $]$

$MakePing \triangleq [$
 $\quad header \mapsto [magic \mapsto 619259748, command \mapsto "ping"],$
 $\quad payload \mapsto [nonce \mapsto 0]$
 $]$

$MakePong \triangleq [$
 $\quad header \mapsto [magic \mapsto 619259748, command \mapsto "pong"],$
 $\quad payload \mapsto [nonce \mapsto 0]$
 $]$

$MakeInv(blocks) \triangleq [$
 $\quad header \mapsto [magic \mapsto 619259748, command \mapsto "inv"],$
 $\quad payload \mapsto [$
 $\quad\quad count \mapsto 1,$
 $\quad\quad inventory \mapsto \langle [type \mapsto "MSG_BLOCK", hash \mapsto Cardinality(blocks)] \rangle$
 $\quad]$
 $]$

$MakeGetHeaders(blocks) \triangleq [$
 $\quad header \mapsto [magic \mapsto 619259748, command \mapsto "getheaders"],$
 $\quad payload \mapsto [$
 $\quad\quad version \mapsto 170100,$
 $\quad]$
 $]$

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    hashCount      ↦ 1,
    blockLocatorHashes ↦  $\langle \text{Cardinality}(\text{blocks}) \rangle$ ,
    hashStop       ↦ 0
  ]
]

MakeHeaders(blocks)  $\triangleq$  [
  header ↦ [magic ↦ 619259748, command ↦ "headers"],
  payload ↦ [
    count ↦ 1,
    headers ↦  $\langle [version \mapsto 170140, prev\_block \mapsto \text{Cardinality}(\text{blocks}), merkle\_root \mapsto 0, bits \mapsto 0, nonce \mapsto 0] \rangle$ 
  ]
]

MakeGetData(blocks)  $\triangleq$  [
  header ↦ [magic ↦ 619259748, command ↦ "getdata"],
  payload ↦ [
    count ↦ 1,
    inventory ↦  $\langle [type \mapsto \text{"MSG\_BLOCK"}, hash \mapsto \text{Cardinality}(\text{blocks})] \rangle$ 
  ]
]

MakeReject(rejected_command)  $\triangleq$  [
  header ↦ [magic ↦ 619259748, command ↦ "reject"],
  payload ↦ [message ↦ rejected_command]
]

MakeBlock(blocks)  $\triangleq$  [
  header ↦ [magic ↦ 619259748, command ↦ "block"],
  payload ↦ [
    version ↦ 70015,
    prev_block ↦  $\text{Cardinality}(\text{blocks})$ ,
    merkle_root ↦ 0,
    bits ↦ 0,
    nonce ↦ 0,
    transactions ↦  $\langle \rangle$ 
  ]
]

```
